

SoftPOS Specification

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Changelog

Version	Description
1.0	Initial Version
1.1	Updated notification and added API Specification
1.2	Updated inquiry specs
1.3	Update SoftPOS Specification

1. Overview

This document represents a transaction result notification sent by Acquirer to 2C2P Callback System once a transaction has been processed.

The purpose of this JSON payload is to synchronize transaction states between the acquirer platform and 2C2P system by providing a standardized data format containing transaction metadata, payment results, and reconciliation references.

This ensures that the all system can:

- Record the final transaction status (APPROVED, VOIDED, DECLINED, etc.)
- Update internal order states (e.g., mark order as paid, refunded, or voided)
- Reconcile settlements using the tranId, rrn, and batchNo
- Display accurate payment details such as masked PAN, card scheme, and approval code

2. Data Structure

Attribute	Description
Field Name	The JSON key.
Data Type	string, integer, boolean, object, array, etc.
Required / Optional	Whether the field must appear in the request.
Format / Constraints	Examples: ISO 8601 date, max length 32 chars, regex pattern.
Allowed Values	Enumerations such as APPROVED, DECLINED, VOIDED.
Example	Example value for clarity.

3. Specification

3.1 HTTP Header

Each callback request includes an authentication header used to verify the origin and integrity of the message.

Header Key	Description
webhook-jwt	A JWT token used to verify the authenticity of the callback request. The token is signed using the ES256 (Elliptic Curve Digital Signature Algorithm) with the acquirer’s private key. 2C2P will verify the token using the corresponding public key provided by the acquirer.

3.1.1 JWT Specification

3.1.1.1 Algorithm

- **Signing Algorithm:** ES256
- **Public Key Format:** X.509 certificate (.cer / .pem)

3.1.1.2 Key Exchange

- Acquirer generates an **ECDSA** key.
- The **private key** is securely held by the acquirer and used to sign outgoing JWTs.
- The **public key certificate (X.509 format)** is shared with 2C2P through a secure channel (e.g., HTTPS portal or encrypted email).

Example:

```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

The JWT payload should match exactly as the notification body below.

3.2 Body

Root Object

Field	Type	Required	Description
tranId	string	✔	Unique transaction identifier. Example: M2067524593572634625
tranStatus	string	✔	Status: APPROVED, DECLINED, VOIDED
tranType	string	✔	Type: SALE, REFUND, AUTH, VOID, CAPTURE
paymentMethod	string	✔	Card brand: VISA, MASTERCARD, AMEX, etc.
posReference	string	✔	POS reference / payment token. Used for TTOD detection
issCountryCode	string	✔	ISO numeric issuer country. Example: 0702 (Singapore)
linkedTranId	string	✔	Linked transaction ID (same as t ranId for original txns)
deviceAlias	string	✔	Device identifier. Example: UNKNOWN
amount	object	✔	Transaction amount object
cardPan	string	✔	Encrypted Card PAN (Base64, AES-128-GCM)
extraData	object	✔	Supplementary metadata object

amount Object

Field	Type	Required	Description
currency	string (ISO 4217)	✔	Currency code of the transaction (e.g., SGD, USD).
value	integer	✔	Transaction amount in the smallest currency unit (e.g., cents). Example: 590 = 5.90 SGD.

extraData Object

Field	Type	Required	Description
merchantId	string	✔	2C2P Merchant ID
paymentToken	string	✖	Optional: Payment Token from PGW
subMid	string	✔	2C2P Merchant ID
subTid	string	✔	Acquirer-assigned merchant ID.

3.3 Sample Notification

3.3.1 Sale

```
{
  "tranId": "M2067524593572634625",
  "tranStatus": "APPROVED",
  "tranType": "SALE",
```

```
"paymentMethod": "MASTERCARD",
"posReference": "1781771250027",
"linkedTranId": "M2067524593572634625",
"deviceAlias": "UNKNOWN",
"amount": {
  "currency": "SGD",
  "value": 130
},
"extraData": {
  "subMid": "702702000004178",
  "subTid": "DCSSP003",
  "jwtSecret": "32D3A904F82E3EE7B502F5A0BB02499B154D2990F8A8113E2C5C071A0AEA9606"
}
}
```

3.3.2 Void

```
{
  "tranId": "M2067524593572634625",
  "tranType": "VOID",
  "tranStatus": "VOIDED",
  "approvalCode": "838601",
  "amount": {
    "currency": "SGD",
    "value": 130
  },
  "posReference": "1781771250027",
  "linkedTranId": "M2067524593572634625",
  "deviceAlias": "UNKNOWN",
  "paymentMethod": "MASTERCARD",
  "extraData": {
    "subMid": "702702000004178",
    "subTid": "DCSSP003",
    "jwtSecret": "..."
  }
}
```

3.3.3 Refund

```
{
  "tranId": "M2067524593572634999",
  "linkedTranId": "M2067524593572634625",
  "tranType": "REFUND",
  "tranStatus": "APPROVED",
  "approvalCode": "R54321",
  "amount": {
    "currency": "SGD",
    "value": 130
  },
  "posReference": "1781771260027",
}
```

```
"deviceAlias": "UNKNOWN",
"paymentMethod": "MASTERCARD",
"extraData": {
  "subMid": "702702000004178",
  "subTid": "DCSSP003",
  "jwtSecret": "..."}
}
```

3.3.4 Capture

```
{
  "tranId": "M2067524593572634999",
  "linkedTranId": "M2067524593572634625",
  "tranType": "AUTH_COMP",
  "tranStatus": "APPROVED",
  "approvalCode": "838601",
  "amount": {
    "currency": "SGD",
    "value": 130
  },
  "posReference": "1781771260027",
  "deviceAlias": "UNKNOWN",
  "paymentMethod": "MASTERCARD",
  "extraData": {
    "subMid": "702702000004178",
    "subTid": "DCSSP003",
    "jwtSecret": "..."}
}
```

3.4 Enumerations

tranType

Specifies the type of transaction performed.

Value	Description
SALE	Standard purchase transaction.
REFUND	Refund issued to a previous sale.
AUTH	Authorization-only transaction (capture may occur later).
AUTH_COMP	Complete an authorization

status

Identifies the status of the transaction Type

Value	Description
PROCESSING	Use to indicate that a transaction is still being processed
APPROVED	Use to indicate that a transaction has been approved
DECLINED	Use to indicate that a transaction has been declined
VOIDED	Voided transaction

paymentMethod

Identifies the card brand or payment channel used.

Value	Description
- VISA - MASTERCARD - UNIONPAY - AMEX - JCB - DISCOVER - DINERS	Card schemes
- WECHAT - ALIPAY_PLUS - SHOPEE_PAY_QR - GRAB_PAY_QR - PROMPT_PAY_QR	QR payment channels.
UNKNOWN	Payment method not identified.

4. API Interface

4.1.1 Inquiry Request

GET request to {base URL}/order/reference/{transactionId}

No body in Inquiry request

Header

Key	Value
Authorization	Bearer {token}

4.1.2 Inquiry Response

```
{
  "msg": "SUCCESS",
  "code": 0,
  "data": {
    "tranId": "M1934506507166126081",
```

```
    "approvalCode": "150107",
    "cvmPerformed": "NO_CVM",
    "deviceAlias": "UNKNOWN",
    "extraData": "{ \"paymentToken\": \"\", \"subMid\": \"\", \"acqMid\": \"\", \"subTid\": \"\", \"clientIp\": \"\", \"merchantName\": \"\", \"tranStatus\": \"APPROVED\", \"tranType\": \"SALE\", \"createdAt\": \"2025-06-16T07:01:07+0000\", \"trace\": \"019439\", \"providerReference\": \"160625152130\", \"callbackUrl\": \"\", \"issCountryCode\": \"0702\", \"entryMode\": \"NFC\", \"updatedAt\": \"2025-06-16T07:01:09+0000\", \"amount\": { \"currency\": \"SGD\", \"value\": 1000 }, \"batchNo\": \"003500\", \"linkedTranId\": \"M1934506507166126081\", \"merchantAddr\": \"\", \"rrn\": \"250616150107\", \"accountMasked\": \"435778*****1111\", \"sdkId\": \"59e2700dcc1227ff\", \"paymentMethod\": \"VISA\", \"hostMessageFormat\": \"CARD_NOW\", \"posReference\": \"160625152130\", \"actions\": [] }
  }
```

Inquiry Response Root Object

Root Object

Field	Type	Required	Description	Example Value
msg	string		Response message	"SUCCESS"
code	integer		Response code (0 = success)	0
data	object		Transaction data object	—

Data Object (within root)

Field	Type	Required	Description	Example Value
tranId	string		Unique transaction identifier	"M1934506507166126081"
approvalCode	string		Issuer approval code	"150107"
cvmPerformed	string		Card verification method	"NO_CVM"
deviceAlias	string		Device identifier	"UNKNOWN"

extraData	string	✔	JSON string with supplementary data	"{\\"paymentToken\\":\\"\\",...}"
mcc	string	✘	Merchant category code	""
merchantName	string	✘	Merchant name	""
tranStatus	string	✔	Transaction status	"APPROVED"
tranType	string	✔	Transaction type	"SALE"
createdAt	string	✔	Creation timestamp (ISO 8601)	"2025-06-16T07:01:07+0000"
trace	string	✔	Trace number	"019439"
providerReference	string	✔	Provider reference ID	"160625152130"
callbackUrl	string	✘	Callback URL for notifications	""
issCountryCode	string	✔	Issuer country code	"0702"
entryMode	string	✔	Transaction entry mode	"NFC"
updatedAt	string	✔	Last update timestamp (ISO 8601)	"2025-06-16T07:01:09+0000"
amount	object	✔	Amount details object	—
batchNo	string	✔	Batch number	"003500"
linkedTranId	string	✔	Linked transaction ID	"M1934506507166126081"
merchantAddr	string	✘	Merchant address	""
rrn	string	✔	Retrieval reference number	"250616150107"
accountMasked	string	✔	Masked account number	"435778*****1111"
sdkId	string	✔	SDK identifier	"59e2700dcc1227ff"
paymentMethod	string	✔	Payment method/brand	"VISA"
hostMessageFormat	string	✔	Host message format	"CARD_NOW"
posReference	string	✔	POS reference number	"160625152130"
actions	array	✘	Available actions array	[]

Amount Object

Field	Type	Required	Description	Example Value
currency	string	✔	Currency code (ISO 4217)	"SGD"
value	integer	✔	Amount in smallest currency unit	1000

extraData Object

Field	Type	Required	Description
merchantId	string	✔	2C2P / acquirer merchant identifier.

acquirerMid	string	✔	Merchant ID assigned by the acquiring bank.
acquirerTid	string	✔	Terminal ID assigned by the acquiring bank.

amount Object

Field	Type	Required	Description
currency	string (ISO 4217)	✔	Currency code of the transaction (e.g., SGD, USD).
value	integer	✔	Transaction amount in the smallest currency unit (e.g., cents). Example: 590 = 5.90 SGD

4.2.1 Refund Request

POST request to {base URL}/api/v2/order/refund/{transactionId}

```
{
  "refundAmount": "1.30",
  "refundReason": "2C2P REFUND",
  "adminPwd": "<Base64-encoded admin password>"
}
```

4.2.2 Refund Response

```
{
  "msg": "Success",
  "code": 0,
  "data": {
    "tranId": "tran_01KVD0MJW0WQD467771MAZBWMY",
    "srsTranId": "M2067538506159943681",
    "consumerPaymentDevice": "Card",
    "approvalCode": "172248",
    "cvmPerformed": "NO_CVM",
    "extraData": "{\"4F\":\"A0000000031010\",\"50\":\"5649534120435245444954\",\"82\":\"2000\",\"84\"",
    "accountBin": "483585",
    "accountLast4": "7839",
    "tsi": "0000",
    "mcc": "5999",
    "acptId": "acpt_01JV6H631H6GGCETS825K9ECSP",
    "batchId": "batch_01KVD0MJTNCZNBG8SCJ3R6YV2G",
    "merchantName": "2C2P Sandbox",
    "linkedSrsTranId": "M1944946026604732417",
    "tranType": "AUTH_COMP",
    "tranStatus": "APPROVED",
    "atc": "0068",
    "createdAt": "2026-06-18T17:22:48.326+08:00",
    "acceptanceId": "acpt_01JV6H631H6GGCETS825K9ECSP",
    "trace": "035563",
    "providerReference": "ms_763779522690",
    "issCountryCode": "0702",
  }
}
```

```
"providerMchId": "M1734573330",
"entryMode": "RETRIEVE",
"batchNo": "005879",
"amount": {
  "currency": "SGD",
  "value": "60.00"
},
"primaryMid": "065240116000002",
"appName": "VISA CREDIT",
"linkedTranId": "tran_01K05YAWC2VBGJSAFYBXRTAJ3J",
"merchantAddr": "128 Beach Road #21-04 Guoco Midtown Singapore 189773",
"primaryTid": "90004300",
"rrn": "260618172248",
"tc": "F05D483A865824EA",
"tvr": "0000000000",
"subMid": "702840000000000",
"subTid": "MINESECSOFTPOS01",
"profileId": "prof_01JV6H6315VQAX0GY1RNP5737",
"accountMasked": "483585*****7839",
"sdkId": "f82b48c79c000a5c",
"paymentMethod": "VISA",
"hostMessageFormat": "CARD_NOW",
"aid": "A0000000031010",
"actions": [
  {
    "tranId": "tran_01KVD0MJW0WQD467771MAZBWMY",
    "reason": "HostSuccess",
    "actionType": "AUTH_COMP",
    "createdAt": "2026-06-18T17:22:48.330+08:00",
    "amount": {
      "currency": "SGD",
      "value": "60.00"
    }
  },
  "trace": "035563",
  "actionStatus": "SUCCESS",
  "extraData": "{\"4F\":\"A0000000031010\",\"50\":\"5649534120435245444954\",\"82\":\"20060618172248.330+08:00\"}",
  "requestId": "req_01KVD0MJ05QPVC3Q35YTTQ800W",
  "hostRespCode": "00",
  "actionId": "act_01KVD0MJW0R6WHW6KVFMF407AM",
  "hostRespExtra": {
    "srsTranId": "M2067538506159943681",
    "9F1A": "0840",
    "9F36": "0068",
    "5F34": "01",
    "9F37": "A871E898",
    "linkedSrsTranId": "M1944946026604732417",
    "4F": "A0000000031010",
    "panToken": "",
    "9F35": "21",
    "9F10": "06021203A00000",
    "9F33": "0068C8",
    "5F2A": "0840",
    "tapToOwnDevice": "false",
    "50": "5649534120435245444954",
```

```
"95": "0000000000",
"57": "4835850016137839D281020110000080",
"9F6E": "20700000",
"9A": "250715",
"9F27": "80",
"9B": "0000",
"9C": "00",
"9F03": "000000000000",
"9F26": "F05D483A865824EA",
"9F09": "0096",
"activationCodeAlias": "702840000000000",
"5F28": "0702",
"mchInputReference": "0a806b2361df44f2af1b3f6c",
"linkedOrgActionTrace": "021437",
"9F02": "000000006000",
"9F66": "26804080",
"clientIp": "18.162.119.203",
"82": "2000",
"posReference": "M1944946026604732417",
"9F1E": "5465726D696E616C",
"84": "A0000000031010",
"linkedTranAmountValue": "60.00"
```

```
}
```

```
}
```

```
]
```

```
}
```

```
}
```

Response Root Object

Field	Type	Required	Description
msg	string	✔	Response Message
code	string	✔	Response Code
data	object	✖	Optional Data field

4.3.1 Void Request

POST /order/void/{transactionId}

```
{
  "adminPwd" : "000000"
}
```

4.3.2 Void Response

```
{
  "msg": "Success",
  "code": 0,
  "data": {
    "payAmount": 130,
```

```
    "acquirerMid": "000240724000056",
    "cardExpData": "1230",
    "acquirerTraceNo": "838601",
    "acquirerBatchNo": "002828",
    "cardMaskedPan": "534967*****6992",
    "state": "VOIDED",
    "acquirerTid": "10002965",
    "acquirerInvoiceNo": "M2067524593572634625",
    "transactionId": "M2067524593572634625",
    "refundId": "V2067524593572634625",
    "refundAmount": 130
  }
}
```

Response Root Object

Field	Type	Required	Description
msg	string		Status message
code	integer		Response code (0 = success)
data.state	string		Must be VOIDED
data.acquirerTraceNo	string		Approval/trace number
data.acquirerInvoiceNo	string		Reference number

4.4.1 Capture Request

POST /api/v2/order/capture/{transactionId}

```
{
  "amount": "1.30"
}
```

4.4.2 Capture Response

```
{
  "msg": "Success",
  "code": 0,
  "data": {
    "tranId": "M2067524593572634999",
    "srsTranId": "S2067524593572634999",
    "consumerPaymentDevice": "MASTERCARD",
    "approvalCode": "838601",
    "cvmPerformed": "NO_CVM",
    "accountBin": "534967",
    "accountLast4": "6992",
    "tsi": "2000",
    "mcc": "",
    "acptId": "10002965",
```

```
"batchId": "002828",
"merchantName": "JR Vending Pte Ltd",
"linkedSrsTranId": "S2067524593572634625",
"tranType": "AUTH_COMP",
"tranStatus": "APPROVED",
"atc": "036D",
"createdAt": "1781771260000",
"acceptanceId": "ACC123456",
"trace": "184777",
"providerReference": "PROV123456",
"issCountryCode": "0702",
"providerMchId": "702702000004178",
"entryMode": "NFC",
"batchNo": "002828",
"amount": {
  "currency": "SGD",
  "value": "1.30"
},
"primaryMid": "702702000004178",
"appName": "4445424954204D415354455243415244",
"linkedTranId": "M2067524593572634625",
"merchantAddr": "",
"primaryTid": "DCSSP003",
"rrn": "606180057610",
"tc": "4307A1ABB01814EC",
"tvr": "0000000001",
"subMid": "702702000004178",
"subTid": "DCSSP003",
"profileId": "PROF123",
"accountMasked": "534967*****6992",
"sdkId": "661f7120bec6e20011ef3f37",
"paymentMethod": "MASTERCARD",
"hostMessageFormat": "CARD_NOW",
"aid": "A00000000041010",
"actions": []
}
}
```

Field	Type	Required	Description
msg	string		Status message
code	integer		Response code (0 = success)
data.tranId	string		New capture transaction ID
data.srsTranId	string		SRS transaction ID
data.tranType	string		Must be AUTH_COMP
data.tranStatus	string		Must be APPROVED
data.linkedTranId	string		Original AUTH tranId
data.approvalCode	string		Capture approval code

5. Security and Encryption

5.1 Card Data Encryption

Sensitive fields such as cardPan are encrypted to protect cardholder data in transit.

Parameter	Encryption Method	Key Size	Padding	Output Format
cardPan	AES/GCM/NoPadding	128 bits	None (GCM mode provides authentication)	Base64 encoded string

Acquirer should share key with 2C2P.

5.1.1 Encryption Workflow

1. The acquirer encrypts the plain card PAN using **AES/GCM/NoPadding** with a 128-bit symmetric key.
2. A random **12-byte IV (Initialization Vector)** is generated for each encryption.
3. The resulting ciphertext is combined with the **GCM authentication tag**.
4. The final encrypted data is encoded to **Base64** before being placed in the cardPan field.

Example:

Plain PAN: 5446241234567261

AES Key (128-bit): <Shared secret known between acquirer and merchant>

Encrypted (AES/GCM): 0x04e7a2fcb1d89a1e... (binary)

Base64 Output: BZYUrndJASATL5pJPYiGJrhFrs5S0Zy5+cQN2/1MNpT7p7nMuUIUXUF01w=

5.1.2 Decryption (2C2P Side)

1. Merchant retrieves the **shared AES key** securely provisioned out-of-band.
2. Decode the Base64-encoded cardPan value.
3. Decrypt using AES/GCM/NoPadding with the same 128-bit key and IV.
4. Validate the **GCM authentication tag** to ensure message integrity.

If decryption fails or authentication fails, 2C2P will reject and log the notification as **tampered or invalid**.

6. Error Codes

Code	HTTP Status	Description	Action
0	200 OK	Success	Process normally
401	401 Unauthorized	Authentication failed	Check credentials
10000	400 Bad Request	Invalid Parameter	Check request body fields
10001	404 Not Found	Transaction Not Found	Verify transactionId exists
12010	409 Conflict	Invalid State	Transaction cannot be voided/refunded in current state
12015	409 Conflict	Already Voided	Transaction already voided
12021	409 Conflict	Already Refunded	Transaction already refunded
12102	409 Conflict	Already Captured	AUTH already captured

12203	400 Bad Request	Invalid Refund Amount	Amount exceeds original transaction
12204	400 Bad Request	Refund Expired	Refund window exceeded
12207	400 Bad Request	Partial Refund Limit	Maximum partial refunds reached
12214	400 Bad Request	Refund Error	Other refund validation error
12215	400 Bad Request	Refund Error	Other refund validation error
12220	400 Bad Request	Refund Error	Other refund validation error
12221	400 Bad Request	Refund Error	Other refund validation error
12222	400 Bad Request	Refund Error	Other refund validation error
<i>other</i>	500 Internal Server Error	Unknown Error	Contact acquirer support

No labels